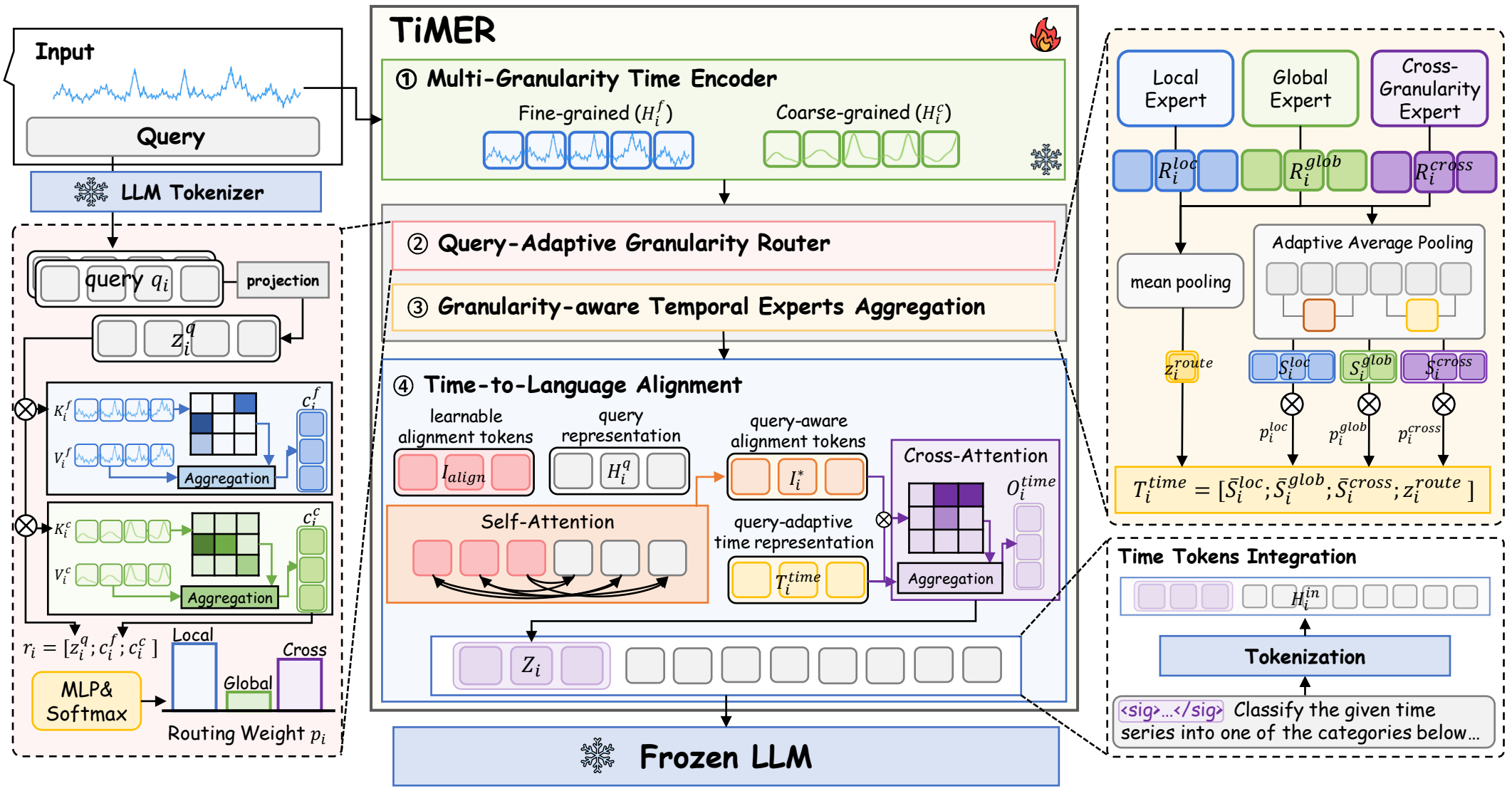

Input
Query
LLM Tokenizer
query q_i
projection
z_i^q
K_i^f
V_i^f
Aggregation
c_i^f
K_i^c
V_i^c
Aggregation
c_i^c
$r_i = [z_i^q; c_i^f; c_i^c]$
MLP& Softmax
Local
Global
Cross
Routing Weight p_i
TiMER
① Multi-Granularity Time Encoder
Fine-grained (H_i^f)
Coarse-grained (H_i^c)
② Query-Adaptive Granularity Router
③ Granularity-aware Temporal Experts Aggregation
④ Time-to-Language Alignment
learnable alignment tokens
I_{align}
query representation
H_i^q
query-aware alignment tokens
I_i^*
Cross-Attention
O_i^{time}
Self-Attention
query-adaptive time representation
T_i^{time}
Aggregation
Z_i
Frozen LLM
Local Expert
Global Expert
Cross-Granularity Expert
R_i^{loc}
R_i^{glob}
R_i^{cross}
mean pooling
Adaptive Average Pooling
z_i^{route}
S_i^{loc}
S_i^{glob}
S_i^{cross}
p_i^{loc}
p_i^{glob}
p_i^{cross}
$T_i^{time} = [\bar{S}_i^{loc}; \bar{S}_i^{glob}; \bar{S}_i^{cross}; z_i^{route}]$
Time Tokens Integration
H_i^{in}
Tokenization
<sig>...</sig> Classify the given time series into one of the categories below...